

TECHNICAL SPECIFICATION



Assessment of power quality – Characteristics of electricity supplied by public networks



THIS PUBLICATION IS COPYRIGHT PROTECTED

Copyright © 2015 IEC, Geneva, Switzerland

All rights reserved. Unless otherwise specified, no part of this publication may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm, without permission in writing from either IEC or IEC's member National Committee in the country of the requester. If you have any questions about IEC copyright or have an enquiry about obtaining additional rights to this publication, please contact the address below or your local IEC member National Committee for further information.

IEC Central Office
3, rue de Varembe
CH-1211 Geneva 20
Switzerland

Tel.: +41 22 919 02 11
Fax: +41 22 919 03 00
info@iec.ch
www.iec.ch

About the IEC

The International Electrotechnical Commission (IEC) is the leading global organization that prepares and publishes International Standards for all electrical, electronic and related technologies.

About IEC publications

The technical content of IEC publications is kept under constant review by the IEC. Please make sure that you have the latest edition, a corrigenda or an amendment might have been published.

IEC Catalogue - webstore.iec.ch/catalogue

The stand-alone application for consulting the entire bibliographical information on IEC International Standards, Technical Specifications, Technical Reports and other documents. Available for PC, Mac OS, Android Tablets and iPad.

IEC publications search - www.iec.ch/searchpub

The advanced search enables to find IEC publications by a variety of criteria (reference number, text, technical committee,...). It also gives information on projects, replaced and withdrawn publications.

IEC Just Published - webstore.iec.ch/justpublished

Stay up to date on all new IEC publications. Just Published details all new publications released. Available online and also once a month by email.

Electropedia - www.electropedia.org

The world's leading online dictionary of electronic and electrical terms containing more than 30 000 terms and definitions in English and French, with equivalent terms in 15 additional languages. Also known as the International Electrotechnical Vocabulary (IEV) online.

IEC Glossary - std.iec.ch/glossary

More than 60 000 electrotechnical terminology entries in English and French extracted from the Terms and Definitions clause of IEC publications issued since 2002. Some entries have been collected from earlier publications of IEC TC 37, 77, 86 and CISPR.

IEC Customer Service Centre - webstore.iec.ch/csc

If you wish to give us your feedback on this publication or need further assistance, please contact the Customer Service Centre: csc@iec.ch.

TECHNICAL SPECIFICATION



Assessment of power quality – Characteristics of electricity supplied by public networks

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

ICS 17.220.99

ISBN 978-2-8322-2547-9

Warning! Make sure that you obtained this publication from an authorized distributor.

CONTENTS

FOREWORD.....	4
INTRODUCTION.....	6
1 Scope.....	7
2 Normative references	8
3 Terms and definitions	9
4 Recommended values for power quality indices.....	16
4.1 General.....	16
4.2 Frequency deviation.....	16
4.3 Supply voltage deviation	16
4.3.1 General	16
4.3.2 Low voltage systems	16
4.3.3 Medium voltage systems.....	17
4.3.4 High voltage systems.....	17
4.4 Voltage unbalance	18
4.5 Flicker.....	18
4.6 Harmonic and interharmonic voltage	18
4.6.1 General	18
4.6.2 Low voltage systems	18
4.6.3 Medium voltage systems.....	19
4.6.4 High voltage systems.....	20
4.7 Voltage dip	21
4.8 Voltage swell	22
4.9 Voltage interruption.....	22
4.10 Mains signalling voltage.....	22
4.11 Rapid voltage change	23
4.12 Transient overvoltage	23
4.12.1 Low voltage systems	23
4.12.2 Medium and High voltage systems.....	24
5 Objectives and methods for power quality assessment	24
5.1 General.....	24
5.2 Site power quality assessment.....	24
5.2.1 General	24
5.2.2 For continuous phenomena.....	25
5.2.3 For discontinuous phenomena (single event)	26
5.3 System aspect power quality assessment	27
5.3.1 General	27
5.3.2 For continuous phenomena.....	27
5.3.3 For discontinuous phenomena (events)	27
Annex A (informative) Examples of PROFILES for Power Quality Specification.....	30
A.1 LV public distribution in European countries.....	30
A.2 LV, MV and HV power supply system in China	31
A.3 Example of a transmission system in Canada	32
A.4 Examples of Profiles in Australia.....	33
Annex B (informative) Example on System Aspect Continuous Disturbance Evaluation.....	34
Annex C (informative) Main Impact of Poor Power Quality.....	35

C.1	Harmonic distortion	35
C.2	Voltage unbalance	35
C.3	Voltage deviation	36
C.4	Frequency deviation.....	36
C.5	Voltage fluctuation	36
C.6	Flicker.....	36
C.7	Voltage dip (or Voltage sag).....	36
C.8	Transient overvoltages.....	36
Annex D (informative) Power Quality Issues Related to Distributed Generation and Micro-grids.....		37
D.1	Voltage deviation	37
D.2	Harmonics	37
D.3	DG magnetic bias (DC current injection)	37
D.4	Voltage fluctuation and flicker	37
D.5	High frequency conducted disturbances.....	38
Annex E (informative) Methods to Maintain and Improve Power Quality		39
E.1	Voltage deviation	39
E.2	Harmonics	39
E.3	Flicker.....	40
E.4	Unbalance	40
E.5	Voltage dip/swell/short time interruption.....	40
Annex F (informative) Relation between Power quality and EMC		41
Bibliography.....		43
Figure 1 – Signal voltages recommended values in percent of U_N used in public LV networks (or U_C in public MV networks)		23
Figure 2 – An example for illustrating voltage THD assessment result trends.....		26
Figure 3 – An example showing information of single event assessment.....		27
Figure F.1 – Application points in a LV system (example)		42
Figure F.2 – Relation between disturbance levels (Schematic significance only)		42
Table 1 – Classification of electromagnetic phenomena addressed by power quality indices.....		8
Table 2 – Flicker severity P_{It} recommended values.....		18
Table 3 – Recommended values of individual harmonic voltages at the low voltage supply terminals for orders up to 50 given in percent of the fundamental voltage U_1		19
Table 4 – Recommended values of individual harmonic voltages at the medium voltage supply terminals for orders up to 50 given in percent of the fundamental voltage U_1		20
Table 5 – Indicative values of individual harmonic voltages at the high voltage supply terminals given in percent of the fundamental voltage U_1		21
Table 6 – Site power quality assessment methods		25
Table 7 – Example of single event assessment.....		27
Table 8 – List of individual events measured at a single monitoring site.....		28
Table 9 – SARFI-X indices coming out of Table 8		28
Table 10 – Magnitude – duration table format		29
Table B.1 – Listing of System Power Quality Evaluation		34

INTERNATIONAL ELECTROTECHNICAL COMMISSION

**ASSESSMENT OF POWER QUALITY – CHARACTERISTICS OF
ELECTRICITY SUPPLIED BY PUBLIC NETWORKS**

FOREWORD

- 1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as "IEC Publication(s)"). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
- 2) The formal decisions or agreements of IEC on technical matters express, as nearly as possible, an international consensus of opinion on the relevant subjects since each technical committee has representation from all interested IEC National Committees.
- 3) IEC Publications have the form of recommendations for international use and are accepted by IEC National Committees in that sense. While all reasonable efforts are made to ensure that the technical content of IEC Publications is accurate, IEC cannot be held responsible for the way in which they are used or for any misinterpretation by any end user.
- 4) In order to promote international uniformity, IEC National Committees undertake to apply IEC Publications transparently to the maximum extent possible in their national and regional publications. Any divergence between any IEC Publication and the corresponding national or regional publication shall be clearly indicated in the latter.
- 5) IEC itself does not provide any attestation of conformity. Independent certification bodies provide conformity assessment services and, in some areas, access to IEC marks of conformity. IEC is not responsible for any services carried out by independent certification bodies.
- 6) All users should ensure that they have the latest edition of this publication.
- 7) No liability shall attach to IEC or its directors, employees, servants or agents including individual experts and members of its technical committees and IEC National Committees for any personal injury, property damage or other damage of any nature whatsoever, whether direct or indirect, or for costs (including legal fees) and expenses arising out of the publication, use of, or reliance upon, this IEC Publication or any other IEC Publications.
- 8) Attention is drawn to the Normative references cited in this publication. Use of the referenced publications is indispensable for the correct application of this publication.
- 9) Attention is drawn to the possibility that some of the elements of this IEC Publication may be the subject of patent rights. IEC shall not be held responsible for identifying any or all such patent rights.

The main task of IEC technical committees is to prepare International Standards. In exceptional circumstances, a technical committee may propose the publication of a technical specification when

- the required support cannot be obtained for the publication of an International Standard, despite repeated efforts, or
- the subject is still under technical development or where, for any other reason, there is the future but no immediate possibility of an agreement on an International Standard.

Technical specifications are subject to review within three years of publication to decide whether they can be transformed into International Standards.

IEC TS 62749, which is a technical specification, has been prepared by IEC technical committee 8: System aspects of electrical energy supply.

The text of this technical specification is based on the following documents:

DTS	Report on voting
8/1363/DTS	8/1381/RVC

Full information on the voting for the approval of this technical specification can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

The committee has decided that the contents of this publication will remain unchanged until the maintenance result date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- transformed into an International standard,
- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

IMPORTANT – The 'colour inside' logo on the cover page of this publication indicates that it contains colours which are considered to be useful for the correct understanding of its contents. Users should therefore print this document using a colour printer.

INTRODUCTION

The description of ELECTRICITY is of fundamental importance within electricity supply systems. In general, its characteristics depend less on its generation than on the way in which it is transported by networks and being used by the equipment of the multiple users. Faults or other events such as short-circuit and lightning strikes occurring within users' installations or public networks also disturb or degrade it.

There is a need for a common set of power quality indices and measurement methods in order to allow different system operators to measure and report power quality in a consistent manner.

Regarding the limits or levels of power quality, the situation differs. Historically, the electrical systems in different countries/regions have been designed in different ways to cater for national/regional variations like different geographic, climatic or commercial conditions, etc. It is thus essential that any set of internationally agreed power quality limits or levels also recognize these differences which depends namely on the system configuration, the transfer characteristics between the different voltage levels (attenuation or amplification), the actual disturbance levels on the system, etc.

Also, the level of power quality is not absolute rather it depends on the price that clients are willing to pay for it. Optimizing power quality should be carried out in a cost-effective manner in that if NETWORK USERS expect power quality to be an intrinsic characteristic of the product they also want it at the lowest price.

This is why some of the objectives recommended hereafter allow for a range of values, or options, while still ensuring the coordination of disturbance levels between different parts of the system or voltage levels.

Then, the requirements to be applied can be expressed by the association of the IEC Power Quality framework from the normative part of this Technical Specification and PROFILES. Examples of profiles are given in Annex A.

Nowadays, Smart Grid construction and massive deployment of renewable energy sources increase the complexity of power quality management.

ASSESSMENT OF POWER QUALITY – CHARACTERISTICS OF ELECTRICITY SUPPLIED BY PUBLIC NETWORKS

1 Scope

This Technical Specification specifies the expected characteristics of electricity at the SUPPLY TERMINALS of public low, medium and high voltage, 50 Hz or 60 Hz, networks.

NOTE 1 The boundaries between the various voltage levels may be different for different countries/regions. In the context of this TS, the following terms for system voltage are used:

- low voltage (LV) refers to $U_N \leq 1 \text{ kV}$;
- medium voltage (MV) refers to $1 \text{ kV} < U_N \leq 35 \text{ kV}$;
- high voltage (HV) refers to $35 \text{ kV} < U_N \leq 230 \text{ kV}$;

NOTE 2 Because of existing network structures, in some countries/regions, the boundary between medium and high voltage can be different.

Most of the recommendations for power quality at the SUPPLY TERMINALS are expressed as POWER QUALITY INDICES that describe the manner in which the characteristics of electricity vary. Such variations may appear random in time, with reference to any specific supply terminal, and random in location, with reference to any given instant of time. As such, the POWER QUALITY INDICES are based on the occurrence of the applicable electromagnetic phenomena:

- continuous phenomena, i.e. deviations from the nominal value that occur continuously over time. Such phenomena occur mainly due to load pattern, changes of load, non-linear loads or distributed generation, and
- discontinuous phenomena or events, i.e. sudden and significant deviations from normal or desired wave shape which typically occur due to unpredictable events (e.g. faults) or external causes (e.g. weather conditions).

The power quality indices and the recommended values are intended to be used as technical reference for regulatory purposes (e.g. in NETWORK CODES) or for contracts between network operator and network user (e.g. part of a CONNECTION AGREEMENT).

Power quality requirements combine the obligations of NETWORK OPERATORS with the requirements of equipment or installations on the electromagnetic environment. It is worth noting however, that the requirements of equipment or installations on the electromagnetic environment also include emission aspects that are addressed in other IEC standards (see Clause 2 and Annex F).

NOTE 3 Network operators are in charge of developing and operating the electricity supply system taking into account at the same time:

- provision of adequate conditions for equipment, installations or other networks connected to their network;
- avoidance of unnecessary costs.

NOTE 4 In many countries/regions, requirements concerning the essential characteristics of electricity at supply terminals of public networks are set, or controlled, by National/Regional Regulatory Authorities.

In some cases, additional requirements or differences in requirements can be agreed by terms of a contract (usually a CONNECTION AGREEMENT) between an individual NETWORK USER and the network operator. Such a contract is most likely to arise for network users with relatively large electricity demand, supplied from the MV or HV network, or having power quality sensitive load. It may also arise in sparsely populated or difficult terrain, such as mountain regions, where distribution costs are high. In such an area a network user may be willing to accept a connection, at lower cost, which does not entirely comply with the power quality standards.

NOTE 5 The quality indices and the recommended values appropriately cover the vast majority of locations under acceptable economic conditions, despite the differences in situations, provided that:

- for mass-market products, emission requirements in standards such as IEC 61000-3-2, 3-3, 3-11 and/or 3-12 are regularly and appropriately updated to take into account the development of markets and changes in technologies;
- for large installations, emission levels are effectively controlled, e.g. through connection agreement (Annex E lists some methods to improve power quality);
- network operators make use of appropriate methodologies and engineering practices, e.g. based on PLANNING LEVELS and IEC TR 61000-3-6, 3-7, 3-13 and/or 3-14.

This Technical Specification applies to the phenomena listed in Table 1.

Table 1 – Classification of electromagnetic phenomena addressed by power quality indices

Continuous phenomena	Discontinuous phenomena – Events
FREQUENCY DEVIATION	SUPPLY INTERRUPTION
SUPPLY VOLTAGE DEVIATION	VOLTAGE DIP
VOLTAGE UNBALANCE	VOLTAGE SWELL
HARMONIC VOLTAGE	TRANSIENT OVERVOLTAGE
INTERHARMONIC VOLTAGE	RAPID VOLTAGE CHANGE
FLICKER (VOLTAGE FLUCTUATION)	
MAINS SIGNALLING VOLTAGES	

NOTE 6 Specification of related measurement methods can be found in IEC 61000-4-30, *EMC – Testing and measurement techniques – Power Quality measurement methods*.

NOTE 7 Specification of the performance of related measuring instruments can be found in IEC 62586, *Power quality measurement in power supply systems*.

While power quality is related to EMC in a number of ways, especially because compliance with power quality requirements depends on the control of cumulative effect of electromagnetic emission from all/multiple equipment and/or installations, this Technical Specification is not an EMC publication (see also Annex F).

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60038, *IEC standard voltages*

IEC 60364-4-44, *Low-voltage electrical installations – Part 4-44: Protection for safety – Protection against voltage disturbances and electromagnetic disturbances*

IEC 60364-5-53, *Electrical installations of buildings – Part 5-53: Selection and erection of electrical equipment – Isolation, switching and control*

IEC 61000-2-2, *Electromagnetic compatibility (EMC) – Part 2-2: Environment – Compatibility levels for low-frequency conducted disturbances and signalling in public low-voltage power supply systems*

IEC TR 61000-2-8, *Electromagnetic compatibility (EMC) – Part 1-8: Environment – Voltage dips and short interruptions on public electric power supply systems with statistical measurement results*

IEC 61000-2-12, *Electromagnetic compatibility (EMC) – Part 2-12: Environment – Compatibility levels for low-frequency conducted disturbances and signalling in public medium-voltage power supply systems*

IEC TR 61000-2-14, *Electromagnetic compatibility (EMC) – Part 2-14: Environment – Overvoltages on public electricity distribution networks*

IEC 61000-3-2, *Electromagnetic compatibility (EMC) – Part 3-2: Limits – Limits for harmonic current emissions (equipment input current ≤ 16 A per phase)*

IEC 61000-3-3, *Electromagnetic compatibility (EMC) – Part 3-3: Limits – Limitation of voltage changes, voltage fluctuations and flicker in public low-voltage supply systems, for equipment with rated current ≤ 16 A per phase and not subject to conditional connection*

IEC TR 61000-3-6, *Electromagnetic compatibility (EMC) – Part 3-6: Limits – Assessment of emission limits for the connection of distorting installations to MV, HV and EHV power systems*

IEC TR 61000-3-7, *Electromagnetic compatibility (EMC) – Part 3-7: Limits – Assessment of emission limits for the connection of fluctuating load installations to MV, HV and EHV power systems*

IEC 61000-3-11, *Electromagnetic compatibility (EMC) – Part 3-11: Limits – Limitation of voltage changes, voltage fluctuations and flicker in public low-voltage supply systems – Equipment with rated current ≤ 75 A and subject to conditional connection*

IEC 61000-3-12, *Electromagnetic compatibility (EMC) – Part 3-12: Limits – Limits for harmonic currents produced by equipment connected to public low-voltage systems with input current >16 A and ≤ 75 A per phase*

IEC TR 61000-3-13, *Electromagnetic compatibility (EMC) – Part 3-13: Limits – Assessment of emission limits for the connection of unbalanced installations to MV, HV and EHV power systems*

IEC TR 61000-3-14, *Electromagnetic compatibility (EMC) – Part 3-14: Limits – Assessment of emission limits for the connection of disturbing installations to LV power systems*

IEC 61000-4-7:2009, *Electromagnetic compatibility (EMC) – Part 4-7: Testing and measurement techniques – General guide on harmonics and interharmonics measurements and instrumentation, for power supply systems and equipment connected thereto*

IEC 61000-4-15, *Electromagnetic compatibility (EMC) – Part 4-15: Testing and measurement techniques – Flickermeter – Functional and design specifications*

IEC 61000-4-30:2008, *Electromagnetic compatibility (EMC) – Part 4-30: Testing and measurement techniques – Power quality measurement methods*

IEC 62586-1, *Power quality measurement in power supply systems – Part 1: Power quality instruments (PQI)*

IEC 62586-2, *Power quality measurement in power supply systems – Part 2: Functional tests and uncertainty requirements*

3 Terms and definitions

For the purpose of this Technical Specification, the following terms and definitions apply.

NOTE Terms are listed in alphabetical order.

3.1

code (in electric power system)

collection of rules concerning rights and duties of the parties involved in a certain part of the electric power system

Note 1 to entry: For example: grid code, distribution code.

[SOURCE: IEC 60050-617:2009, 617-03-03]

3.2

connection agreement

agreement entered between the system operator and a system user which governs the procedure and conditions for connection

[SOURCE: IEC 60050-617:2009, 617-04-03]

3.3

declared supply voltage

U_c (abbreviation)

supply voltage U_c agreed by the network operator and the network user

Note 1 to entry: Generally declared supply voltage U_c is the nominal voltage U_N but it may be different according to the agreement between the network operator and the network user.

3.4

electricity

set of the phenomena associated with electric charges and electric currents

[SOURCE: IEC 60050-121:1998, 121-11-76]

Note 1 to entry: In the context of electric power systems, electricity is often described as a product with particular characteristics.

3.5

electromagnetic compatibility

ability of an equipment or system to function satisfactorily in its electromagnetic environment without introducing intolerable electromagnetic disturbances to anything in that environment

[SOURCE: IEC 60050-161:1990, 161-01-07]

3.6

(electromagnetic) compatibility level

specified electromagnetic disturbance level used as a reference level for coordination in the setting of emission and immunity limits

Note 1 to entry: By convention, the compatibility level is chosen so that there is only a small probability that it will be exceeded by the actual disturbance level.

3.7

flicker

impression of unsteadiness of visual sensation induced by a light stimulus whose luminance or spectral distribution fluctuates with time

Note 1 to entry: Voltage fluctuation cause changes of the luminance of lamps which can create the visual phenomenon called flicker. Above a certain threshold, flicker becomes annoying. The annoyance grows very rapidly with the amplitude of the fluctuation. At certain repetition rates, even very small amplitudes can be annoying.

Note 2 to entry: For the time being, flicker is qualified based on incandescent lamp's behavior.

[SOURCE: IEC 60050-161:1990, 161-08-13, modified (addition of Notes to entry)]

3.8

flicker severity

intensity of flicker annoyance evaluated by the following quantities:

- short term severity (P_{st}) measured over a period of ten minutes;
- long term severity (P_{lt}) calculated from a sequence of 12 P_{st} -values over a two hour interval, according to the following expression:

$$P_{lt} = \sqrt[3]{\sum_{i=1}^{12} \frac{P_{sti}^3}{12}}$$

Note 1 to entry: For details of P_{st} and P_{lt} , see IEC 61000-4-15.

3.9**frequency deviation**

difference between power supply frequency ($f_{H,1}$) and nominal frequency (f_N)

3.10**group total harmonic distortion**

THDG (abbreviation)

THDG_Y(symbol)

ratio of the r.m.s. value of the harmonic groups ($Y_{g,h}$) to the r.m.s. value of the group associated with the fundamental ($Y_{g,1}$):

$$THDG_Y = \sqrt{\sum_{h=2}^{h_{\max}} \left(\frac{Y_{g,h}}{Y_{g,1}} \right)^2}$$

Note 1 to entry: The symbol Y is replaced, as required, by the symbol I for currents or by the symbol U for voltages.

[SOURCE: IEC 61000-4-7:2009, 3.3]

3.11**harmonic frequency**

$f_{H,h}$ (abbreviation)

frequency which is an integer multiple of the power supply (fundamental) frequency

[SOURCE: IEC 61000-4-7:2009, 3.2.1, modified (removal of formula and Note to entry)]

3.12**harmonic order**

h (abbreviation)

(integer) ratio of a harmonic frequency ($f_{H,h}$) to the power supply frequency ($f_{H,1}$)

3.13**harmonic ratio**

HR (abbreviation)

ratio of individual harmonic order component (U_h or I_h) to the fundamental component (U_1 or I_1)

3.14**mains signalling voltage**

signal superimposed on the supply voltage for the purpose of transmission of information in the public supply network and to network users' premises

Note 1 to entry: Three types of signals in the public supply network can be classified:

- *ripple control signals*: superimposed sinusoidal voltage signals in the frequency range 110 Hz to 3 000 Hz;
- *power-line-carrier signals*: superimposed sinusoidal voltage signals in the frequency range 3 kHz to 148,5 kHz;
- *mains marking signals*: superimposed short time alterations (transients) at selected points of the voltage waveform.

3.15**network operator****system operator**

party responsible for safe and reliable operation of a part of the electric power system in a certain area and for connection to other parts of the electric power system

[SOURCE: IEC 60050-617:2009, 617-02-09]

3.16**nominal frequency**

f_N (abbreviation)

value of frequency used to designate or identify a system

3.17

nominal system voltage

U_N (abbreviation)

value of voltage used to designate or identify a system

[SOURCE: IEC 60050-601:1985, 601-01-21, modified (addition of abbreviation, removal of "suitable approximate" from beginning of definition)]

3.18

normal operating conditions (of a public electricity supply system)

operating conditions of a public electricity supply system typically including all generation variations, load variations and reactive compensation or filter states (e.g. shunt capacitor states), planned outages and planned arrangements during maintenance and construction work, non-ideal operating conditions and normal contingencies under which the considered system has been designed to operate

Note 1 to entry: Normal system operating conditions typically exclude exceptional situations such as: conditions arising as a result of a fault or a combination of faults beyond that planned for under the system security standard, unavoidable circumstances (for example: force majeure, exceptional weather conditions and other natural disasters, acts by public authorities, industrial actions), cases where NETWORK USERS significantly exceed their emission limits or do not comply with the connection requirements, and temporary generation or supply arrangements adopted to maintain supply to NETWORK USERS during maintenance or construction work, where otherwise supply would be interrupted.

[SOURCE: IEC TR 61000-3-6:2008, 3.14, modified ("of a public electricity supply system" added to term and definition)]

3.19

percentile value

$U_{x\%}$ (symbol)

value such that x percent ($x\%$) of measurements are smaller than or equal to that value, over a given period

3.20

planning level

level of a particular disturbance in a particular environment, adopted as a reference value for the limits to be set for the emissions from the installations in a particular system, in order to co-ordinate those limits with all the limits adopted for equipment and installations intended to be connected to the power supply system

Note 1 to entry: Planning levels are considered internal quality objectives to be specified at a local level by those responsible for planning and operating the power supply system in the relevant area.

[SOURCE: IEC TR 61000-3-6:2008, 3.16]

3.21

point of common coupling

PCC (abbreviation)

point in a public power supply network, electrically nearest to a particular load, at which other loads are or may be connected

Note 1 to entry: These loads can be either devices, equipment or systems, or distinct network user's installations.

[SOURCE: IEC 60050-161:1990, 161-07-15, modified ("consumer's installation" replaced by "load")]

3.22

supply terminals

point in a distribution network designated as such and contractually fixed, at which electric energy is exchanged between contractual partners

Note 1 to entry: Supply terminals may be different from the boundary between the electricity supply system and the user's own installation or from the metering point.

[SOURCE: IEC 60050-617:2009, 617-04-02, modified Note 1 to entry]

3.23**(power) network user**

party supplying electric power and energy to, or being supplied with electric power and energy from, a transmission system or a distribution system

[SOURCE: IEC 60050-617:2009, 617-02-07]

3.24**power quality**

characteristics of the electricity at a given point on an electrical system, evaluated against a set of reference technical parameters

Note 1 to entry: These parameters might, in some cases, relate to the compatibility between electricity supplied on a network and the loads connected to that network.

Note 2 to entry: In the context of this Technical Specification, power quality refers to supply terminals and focuses on defining the characteristics of the voltage and frequency.

[SOURCE: IEC 60050-617:2009, 617-01-05, modified ("electric current, voltage and frequencies" replaced by "electricity" and Note 2 to entry added)]

3.25**power quality indices**

technical parameters characterizing the quality of electricity, measured at a given point, relevant for the assessment of the quality of the electricity delivered by a network operator

3.26**profile**

specification that supplement a standard by limiting options, in order to serve the needs of users in a geographic area or in an application domain

3.27**r.m.s. value of a harmonic component**

$Y_{H,h}$ (abbreviation)

r.m.s. value of one of the components having a harmonic frequency in the analysis of a non-sinusoidal waveform. For brevity, such a component may be referred to simply as a 'harmonic'

Note 1 to entry: The symbol Y is replaced, as required by the symbol I for currents, by the symbol U for voltages.

Note 2 to entry: For more details, see IEC 61000-4-7:2009.

[SOURCE: IEC 61000-4-7:2009, 3.2.3]

3.28**r.m.s. value of a harmonic group**

$Y_{g,h}$ (abbreviation)

square root of the sum of the squares of the r.m.s. value of a harmonic and the spectral components adjacent to it within the time window, thus summing the energy contents of the neighboring components with that of the harmonic proper

Note 1 to entry: The symbol Y is replaced, as required by the symbol I for currents, by the symbol U for voltages.

Note 2 to entry: For more details, see IEC 61000-4-7:2009.

[SOURCE: IEC 61000-4-7:2009, 3.2.4]

3.29**r.m.s. value of an interharmonic centred subgroup**

$Y_{isg,h}$ (abbreviation)

r.m.s. value of all interharmonic components in the interval between two consecutive harmonic frequencies, excluding frequency components directly adjacent to the harmonic frequencies

Note 1 to entry: The r.m.s. value of the centred subgroup between the harmonic orders h and $h + 1$ is designated as $Y_{isg,h}$; for example, the centred subgroup between $h = 5$ and $h = 6$ is designated as $Y_{isg,5}$.

Note 2 to entry: For more details, see IEC 61000-4-7:2009.

[SOURCE: IEC 61000-4-7:2009, 3.4.4]

3.30

r.m.s. value of an interharmonic component

$Y_{C,i}$ (abbreviation)

r.m.s. value of a spectral component of an electrical signal with a frequency between two consecutive harmonic frequencies

For brevity, such a component may be referred to simply as an 'interharmonic'.

Note 1 to entry: For more details, see IEC 61000-4-7:2009, 3.4.2.

3.31

rapid voltage change

RVC (abbreviation)

quick transition (that may last more than several cycles) in r.m.s. voltage between two steady-state conditions while the voltage stays in-between the thresholds defined for voltage swells and dips (otherwise, it would be considered as a swell or a dip)

RVC is expressed by the relative steady-state voltage change and/or by a maximum relative r.m.s. voltage change aggregated over several cycles.

Note 1 to entry: For more information, see IEC 61000-4-30.

3.32

recommended values

value under which, or values within which, the voltage characteristics should remain in view of providing an acceptable quality of the electricity supply

Note 1 to entry: The characteristics of electricity agreed between network operator and a network user or set by national/regional regulatory authority can be locally optimized.

3.33

reference voltage (for interruptions, voltage dips and voltage swells measurement and evaluation)

value specified as the base on which residual voltage, thresholds and other values are expressed in per unit or percentage terms

3.34

supply voltage

r.m.s. value of the line-to-line or line-to-neutral voltage at a given time at the supply terminal, measured over a given interval

3.35

time aggregation

combination of several sequential values of a given parameter (each determined over identical time intervals) to provide a value for a longer time interval

Note 1 to entry: In this Technical Specification, 3 s value refers to IEC 61000-4-30 150/180-cycle interval aggregation value (150 cycles for 50 Hz nominal or 180 cycles for 60 Hz nominal), 10 min value and 2 h value are also defined in IEC 61000-4-30.

[SOURCE: IEC 61000-4-30:2015, 3.31, modified (modification of Note 1 to entry)]

3.36

total harmonic distortion

THD (abbreviation)

THD_Y (symbol)

ratio of the r.m.s. value of the sum of all the harmonic components ($Y_{H,h}$) up to a specified order (h_{max}) to the r.m.s. value of the fundamental component ($Y_{H,1}$):

$$THD_Y = \sqrt{\sum_{h=2}^{h_{max}} \left(\frac{Y_{H,h}}{Y_{H,1}} \right)^2}$$

Note 1 to entry: The symbol Y is replaced, as required, by the symbol I for currents or by the symbol U for voltages.

Note 2 to entry: For more information, see IEC 61000-4-30:2015, 5.8.1.

3.37

transient overvoltage

voltage surge

transient voltage wave propagating along a line or a circuit and characterized by a rapid increase followed by a slower decrease of the voltage

[SOURCE: IEC 60050-161:1990, 161-08-11]

3.38

voltage deviation

difference between supply voltage (U) to nominal voltage (U_N), often expressed by relative value

Note 1 to entry: In some circumstance, U_N may be replaced by U_C by contract or agreement.

3.39

voltage dip

sudden reduction of the voltage at a point in an electrical system followed by voltage recovery after a short period of time, usually from a few cycles to a few seconds

Note 1 to entry: The starting threshold of voltage dip generally is 90 % of reference voltage.

[SOURCE: IEC 60050-161:1990, 161-08-10, modified (addition of Note 1 to entry)]

3.40

voltage fluctuation

series of voltage changes or a cyclic variation of the supply voltage envelope

Note 1 to entry: For the purpose of this Technical Specification, the reference voltage is the nominal or declared voltage of the supply system.

[SOURCE: IEC 60050-161:1990, 161-08-05, modified (addition of "supply voltage" and Note 1 to entry)]

3.41

voltage short time interruption

disappearance of the supply voltage for a time interval whose duration is between two specified limits

Note 1 to entry: A short time interruption is considered to be a reduction of the supply voltage under the interruption threshold (usually 5 % or 10 % of the reference voltage) of the nominal voltage, with the lower limit of the duration typically a few tenths of a second, and its upper limit typically in the order of 1 min (or, in some cases up to 3 or 5 min).

[SOURCE: IEC 60050-161:1990, 161-08-20, modified Note 1 to entry]

3.42

voltage swell

sudden increase of the voltage at a point in an electrical system followed by voltage recovery after a short period of time, usually from a few cycles to a few seconds

Note 1 to entry: The starting threshold of voltage swell generally is 110 % of reference voltage.

3.43

voltage unbalance

in a polyphase system, a condition in which the magnitudes of the phase voltages or the phase angles between consecutive phases are not all equal (fundamental component)

[SOURCE: IEC 60050-161:1990, 161-08-09, modified ("r.m.s. values" replaced by "magnitudes")]

3.44

voltage unbalance factor

in a three-phase system, the degree of unbalance expressed by the ratio (in per cent) between the r.m.s. values of the negative sequence (or, rarely, of the zero sequence) component and the positive sequence component of voltage

[SOURCE: IEC 60050-604:1987, 604-01-30, modified (addition of "voltage" to term)]

4 Recommended values for power quality indices

4.1 General

Regarding phenomena that occur continuously over time, this Technical Specification provides the recommended or indicative values applicable during normal operating conditions. It considers an observation period of at least one week, e.g. in order to take account for variation in loads.

Measurement of voltage characteristics requires an aggregation time interval for both actual calculation of the voltage and for comparability between results at different points in time. In the case of this Technical Specification, the 10 minutes interval proposed in IEC 61000-4-30 is used for most phenomena.

NOTE 1 In some countries/regions, aggregation time intervals less than 10 minutes are used.

NOTE 2 Voltage fluctuation leading to Long Time Flicker and Mains Signalling has specific observation period and/or aggregation time intervals.

The versatility and adaptability of electricity are managed by assigning some probability factors to the recommended values for some power quality indices. PERCENTILE VALUES are then compared to the recommended values expected to be statistically fulfilled within the observation periods. The related probability should never be less than 95 %.

Flagging concept is used in this Technical Specification according to IEC 61000-4-30. Unless otherwise indicated (e.g. for voltage deviation), the flagged data is excluded in the calculation of percentile values.

Network disturbances corresponding to discontinuous phenomena or events shown in table 1 require a relatively long observation period. Depending on the frequency of their occurrence and the wanted level of statistical accuracy, this period can vary between a single season to several years. These phenomena are mostly unpredictable, which make it very difficult to give useful definite values for the corresponding characteristics. The values given in this Technical Specification for the voltage characteristics associated with such phenomena, i.e. voltage dips/swells, voltage interruptions and rapid voltage changes, shall be interpreted as indicative.

For all phenomena, the measurements are performed according to IEC 61000-4-30.

4.2 Frequency deviation

The frequency shall be maintained within a given deviation from the specified value, 50 Hz or 60 Hz, in order to maintain a stable power system operation.

NOTE In different synchronous areas, different requirements may apply (generally originated from grid codes).

4.3 Supply voltage deviation

4.3.1 General

The voltage deviation values are established according to standard voltages and voltage ranges defined in IEC 60038.

4.3.2 Low voltage systems

For low voltage systems, recommended values are based on NOMINAL VOLTAGE (U_N).

Data flagged by voltage dips and swells shall be included in the calculation of percentile values.

NOTE 1 In low voltage systems, declared and nominal voltage are equal.

NOTE 2 The nominal voltage U_N for public low voltage is either, between line and neutral, or between lines.

Under normal operating conditions, during each period of one week, the 10 minute r.m.s. values of the voltage at the supply terminals (U), excluding the periods of interruptions, should comply with the following conditions:

- voltage percentile $U_{p\%}$ does not exceed $U_N + 10\%$;
- voltage percentile $U_{\beta\%}$ is not lower than $U_N - 10\%$;

p having a value in the range [99, 100] according to national/regional conditions

β having a value of 5 or in the range of [0, 1] according to national/regional conditions

NOTE In some countries/regions, voltage range can be specified asymmetrically with regard to nominal voltage, e.g. $+6\% \sim -14\%$.

- And, if $\beta > 100 - p$, voltage percentile $U_{(100-p)\%}$ is not lower than $U_N - 15\%$.

NOTE The limits of the supply voltage deviation are more restrictive in some countries/regions.

4.3.3 Medium voltage systems

Network users with demands exceeding the capacity of the LV network are generally supplied at nominal voltages above 1 kV. This clause applies to such electricity supplies at nominal voltages up to and including 35 kV.

NOTE Network users may also be supplied at this voltage level to satisfy special requirements or to mitigate conducted disturbances emitted by their equipment.

For medium voltage systems, recommended values are based on DECLARED VOLTAGE (U_C).

Data flagged by voltage dips and swells shall be included in the calculation of percentile values.

Under normal operating conditions, during each period of one week, the 10 minute r.m.s. values of the voltage at the supply terminals (U), excluding the periods of interruption, should comply with the following conditions:

- voltage percentile $U_{99\%}$ does not exceed $U_C + 10\%$;
- voltage percentile $U_{1\%}$ is not lower than $U_C - 10\%$;
- voltage percentile $U_{0\%}$ is not lower than $U_C - 15\%$.

NOTE 1 In some countries/regions, voltage range can be specified asymmetrically with regard to declared voltage.

NOTE 2 In some countries/regions, $U_{100\%}$ doesn't exceed $U_C + 15\%$

4.3.4 High voltage systems

This clause applies to electricity supplies at nominal voltages above 35 kV and not exceeding 230 kV.

NOTE 1 Network users may also be supplied at this voltage level to satisfy special requirements or to mitigate conducted disturbances emitted by their equipment.

NOTE 2 The number of network users supplied directly from HV networks is limited and normally supply voltage is subject to individual contract.

For high voltage systems, recommended values are based on declared voltage (U_C).

Data flagged by voltage dips and swells shall be included in the calculation of percentile values.

Under normal operating conditions, during each period of one week, the 10 minute r.m.s. values of the voltage at the supply terminals (U), excluding the periods of interruption, should comply with the following conditions:

- voltage percentile $U_{99\%}$ does not exceed $U_C + 10\%$;
- voltage percentile $U_{1\%}$ is not lower than $U_C - 10\%$.

4.4 Voltage unbalance

Under normal operating conditions, during each period of one week, the 10 minute value of negative sequence voltage unbalance factor should be less than or equal to the recommended value for 95 % of time or more.

For three phase LV, MV, and HV supply system, negative sequence voltage unbalance factor $\varepsilon(\%)$ recommended value is 2 %.

NOTE In some countries/regions with part single phase or two phase connected network user's installations, unbalance up to 3 % at three phase supply terminals may occur.

4.5 Flicker

Under normal operating conditions, during each period of one week, the flicker severity P_{lt} should be less than or equal to the recommended values defined in Table 2 for 95 % of time or more.

In the case of complaints, the HV and MV limits and appropriate mitigation measures shall be chosen in such a way that at LV the P_{lt} values do not exceed 1.

Table 2 – Flicker severity P_{lt} recommended values

Voltage levels	P_{lt}
LV/MV	1,0
HV	1,0

NOTE 1 The recommended values are based on the effect of voltage fluctuation on classical incandescent lamps. Modern types of lamps may have different behavior.

NOTE 2 These limits take into consideration flicker attenuation from MV to LV and from HV to MV.

4.6 Harmonic and interharmonic voltage

4.6.1 General

Recommended or indicative values are given for individual harmonic voltages, namely HARMONIC RATIO (HR), and for total harmonic distortion (THD).

Resonances may cause higher voltages for an individual harmonic, however these effects should be controlled whenever they may have an impact on the system or equipment.

Recommended values (or limits) will be given hereafter for interharmonic voltages as well. However, these limits are not meant to control flicker effects or interference in ripple control systems, as those interharmonic levels are under consideration, pending more experience.

4.6.2 Low voltage systems

4.6.2.1 Harmonic voltages

Under normal operating conditions, during each period of one week, the voltage percentile $U_{h95\%}$ of the 10 minute r.m.s. values of each individual harmonic voltage should be less than or equal to the values given in Table 3.

Table 3 – Recommended values of individual harmonic voltages at the low voltage supply terminals for orders up to 50 given in percent of the fundamental voltage U_1

Odd harmonics				Even harmonics	
Not multiples of 3		Multiples of 3 ^a			
Order	$U_{h95\%}$	Order	$U_{h95\%}$	Order	$U_{h95\%}$
h	(%)	h	(%)	h	(%)
5	6,0	3	5,0 (6,0)	2	2,0
7	5,0	9	1,5 (3,5)	4	1,5
11	3,5	15	0,5 (2, 0)	6 ... 24	0,75
13	3,0	21	0,5 (1,5)		
17	2,0				
19	1,8				
23	1,5				
25	1,5				
29≤h≤ 49	2,27 × (17/h) – 0,27	27 ≤h≤ 45	0,2	26 ≤h≤ 50	0,25 × (10/h) + 0,25

^a Depending on the type of neutral grounding systems and transformer connections in some countries/regions, more triplen harmonics will flow in neutral conductors and may cause higher harmonic voltages. In these cases, the highest value in brackets in Table 3 should adequately characterize the system harmonic voltages.

The total harmonic distortion (THD) of the supply voltage for harmonic orders up to 50, should be less than or equal to 8 %.

NOTE In some countries/regions, orders up to 40, only, are used to calculate THD, maintaining the same recommended value.

4.6.2.2 Interharmonic voltages

The same statistical approach and same range of values (see Table 3) should apply to harmonic (centred) group voltages as to individual harmonic voltages from the previous section. The group total harmonic distortion (THDG) of the supply voltage should also be less than or equal to 8 %.

In addition, based on a prudent consideration suggested in IEC 61000-2-2, indicative values and indices should be as follows. Under normal operating conditions, during each period of one week, the interharmonic voltage percentile $U_{h95\%}$ of the 10 minute r.m.s. values of each interharmonic centred subgroup voltage should be no higher than those of the adjacent harmonic voltage given in Table 3.

4.6.3 Medium voltage systems

4.6.3.1 Harmonic voltages

Under normal operating conditions, during each period of one week, the voltage percentile $U_{h95\%}$ of the 10 minute r.m.s. values of each individual harmonic voltage should be less than or equal to the values given in Table 4.

Table 4 – Recommended values of individual harmonic voltages at the medium voltage supply terminals for orders up to 50 given in percent of the fundamental voltage U_1

Odd harmonics				Even harmonics	
Not multiples of 3		Multiples of 3 ^a			
Order h	$U_{h95\%}$ (%)	Order h	$U_{h95\%}$ (%)	Order h	$U_{h95\%}$ (%)
5	6,0	3	5,0 (6,0)	2	2,0
7	5,0	9	1,5 (3,5)	4	1,5
11	3,5	15	0,5 (2,0)	6 ... 24	0,75
13	3,0	21	0,5 (1,5)		
17	2,0				
19	1,8				
23	1,5				
25	1,5				
29 ≤ h ≤ 49	2,27 × (17/h) - 0,27	27 ≤ h ≤ 45	0,2	26 ≤ h ≤ 50	0,25 × (10/h) + 0,25
NOTE 1 Where national/regional circumstances make it appropriate, lower limits may be specified than the values given in Table 4.					
NOTE 2 Depending on the type of voltage transformer used, the measurement of high order harmonics may be not reliable; further information is given in IEC 61000-4-30:2008, A.3.3.					
NOTE 3 When end-use equipment is not directly connected to the MV system, lower values for MV may be more appropriate to allow coordination of disturbance levels between low and medium voltage systems.					
^a Depending on the type of neutral grounding systems and transformer connections in some countries/regions, more triplen harmonics will flow in neutral conductors and may cause higher harmonic voltages. In these cases, the highest value in brackets in Table 4 should adequately characterize the system harmonic voltages.					

The total harmonic distortion (THD) of the supply voltage for orders up to 50, should be less than or equal to 8 %.

NOTE In some countries/regions, orders up to 40, only, are used to calculate THD, maintaining the same recommended value.

4.6.3.2 Interharmonic voltages

The same statistical approach and same range of values (see Table 4) should apply to harmonic (centred) group voltages as to individual harmonic voltages from the previous section. The group total harmonic distortion (THDG) of the supply voltage should also be limited under the value of 8 %.

In addition, based on a prudent consideration suggested in IEC 61000-2-12, indicative values and indices should be as follows. Under normal operating conditions, during each period of one week, the interharmonic voltage percentile $U_{h95\%}$ of the 10 minute r.m.s. values of each interharmonic centred subgroup voltage should be no higher than those of the adjacent harmonic voltage given in Table 4.

4.6.4 High voltage systems

4.6.4.1 Harmonic voltages

Under normal operating conditions, during each period of one week, the voltage percentile $U_{h95\%}$ of the 10 minute r.m.s. values of each individual harmonic voltage should be less than or equal to the values given in Table 5.

It is worthwhile mentioning that for HV transmission systems, one must realize the different purpose of voltage quality objectives. In contrast to low or medium voltage systems, quality objectives for harmonics in HV transmission systems are not directly related to their impact on equipment as the end-use equipment is not directly connected at HV. Indeed, these indicative levels are well below the levels that could cause immediate disturbances on the equipment.

The indicative values on transmission systems are aimed at coordinating disturbance levels between different parts of the system or different voltage levels and may serve as indication of holistic network problems which warrant consideration.

Consequently, the indicative values of individual harmonic voltages at the higher voltage level should not be higher than the value for the lower one (MV and LV).

Table 5 – Indicative values of individual harmonic voltages at the high voltage supply terminals given in percent of the fundamental voltage U_1

Odd harmonics				Even harmonics	
Not multiples of 3		Multiples of 3			
Order h	U_h (%)	Order h	U_h (%)	Order h	U_h (%)
5	2,0 – 5,0	3	2,0 – 3,0	2	1,5 – 1,9
7	2,0 – 4,0	9	1,0 – 2,0	4	0,8 – 1
11	1,5 – 3,0			6 .. 12	0,5
13	1,5 – 2,5				

NOTE 1 Limits for individual harmonic voltage of order higher than 13 are not defined due especially to limited accuracy of voltage transformers currently in use on HV systems. For measurement accuracy, an appropriate type of voltage transformer should be used, particularly for the measurement of higher order harmonics.

NOTE 2 Due to the wide range of voltage levels included in HV (35-230 kV) and where national/regional circumstances make it appropriate, limits for intermediate voltage levels may be specified within the range of values given in Table 5.

The total harmonic distortion (THD) of the supply voltage should be limited under a value within 3 % to 6 % (including harmonics for orders up to 50).

NOTE In some countries/regions, orders up to 40, only, are used to calculate THD, maintaining the same recommended value.

4.6.4.2 Interharmonic voltages

The same statistical approach and same range of values (see Table 5) should apply to harmonic (centred) group voltages as to individual harmonic voltages from the previous section. The group total harmonic distortion (THDG) of the supply voltage should also be limited under a value within 3 % to 6 %.

In addition, based on a prudent consideration suggested in IEC 61000-2-12, indicative values and indices should be as follows. Under normal operating conditions, during each period of one week, the interharmonic voltage percentile U_{h95} % of the 10 minute r.m.s. values of each interharmonic centred subgroup voltage should be no higher than those of the adjacent harmonic voltage given in Table 5.

4.7 Voltage dip

Voltage dips typically originate from short-circuits occurring in the public network or in network users' installations. The annual frequency varies greatly depending on the type of supply system and on the point of observation. Moreover, the distribution over the year can be very irregular.

The power quality characteristics of individual events are defined for each individual phase, by residual voltage and duration, despite the specific shape of the r.m.s. voltage variation. More assessment methods are described in 5.2 and 5.3 hereafter.

Where assessments are performed or statistics are collected to be provided to network users or authorities, voltage dips should be classified according to Table 10.

For polyphase measurements, it is recommended that the number of phases affected by each event is detected and stored.

Generally, according to the network user connection, or the concrete situation, line to line or line to neutral voltage shall be considered.

4.8 Voltage swell

Frequently, voltage swell phenomenon may occur to be unpredictable and random. Depending upon the magnitude and duration, voltage swell may affect different types of load differently for the same voltage swell event. Recommended values of voltage swells are still under consideration. More assessment methods are described in 5.2 and 5.3 hereafter.

Where assessment is performed or statistics are collected to be provided to network users or authorities, voltage swells should be classified according to Table 10.

For polyphase measurements, it is recommended that the number of phases affected by each event is detected and stored.

Generally, according to the network user connection, or the concrete situation, line to line or line to neutral voltage shall be considered.

4.9 Voltage interruption

On single-phase systems, a voltage interruption begins when the residual voltage falls under interruption threshold.

On polyphase systems, a voltage interruption begins when the residual voltage of all phases falls under interruption threshold.

Interruption threshold is generally 5 % or 10 % of the reference voltage.

Even when referring only to normal operating conditions, the annual frequency of supply interruptions varies substantially between areas. This is due to, among other things, differences in system layout (e.g. cable systems versus overhead line systems), environmental and climatic conditions.

In most countries/regions, specific continuity of supply indices are established by regulators in order to facilitate benchmarking of the performance of the network operators under their jurisdiction. The indices allow network operators to meet their obligation to routinely report continuity of supply performance.

In the context of this Technical Specification, voltage short time interruptions are mostly addressed. The detection of short time voltage interruptions are referred to in IEC 61000-4-30 and more assessment methods are described in 5.2 and 5.3 hereafter.

Where assessment is performed or statistics are collected to be provided to network users or Authorities, voltage short time interruption should be classified according to Table 10.

4.10 Mains signalling voltage

The public networks may be used by the network operators or network users for the transmission of signals. Three types of systems are considered by standards:

- ripple control systems that are used by electrical utilities in public supply networks, in the range of 100 Hz to 3 kHz, generally below 500 Hz, with signals up to 5 % of U_N under normal circumstances and up to 9 % of U_N in cases of resonance;
- power-line carrier systems used by electrical utilities in public supply networks, in the range 3 kHz to 95 kHz, with allowed signal levels up to 5 % of U_N . These signals are strongly attenuated in the network (>40 dB);
- signalling systems for end-user premises (residential or industrial) in the range of 95 kHz to 148,5 kHz in Europe (ITU region 1), with allowed signal levels up to 0,6 % of U_N or 5 % of U_N , respectively. In some countries/regions the upper frequency is 500 kHz, with allowed signal levels between 2 – 0,6 mV.

Signal voltages recommended values in LV and MV are indicated in Figure 1.

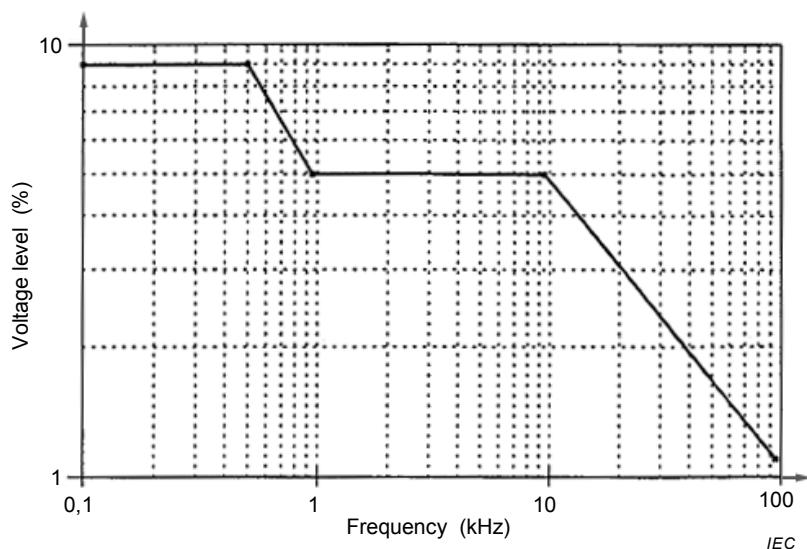


Figure 1 – Signal voltages recommended values in percent of U_N used in public LV networks (or U_C in public MV networks)

NOTE 1 Due to the low resonance frequency of the HV network, no values are given for mains signalling voltages.

NOTE 2 The effective use of mains signalling is challenged by the presence of harmonic, interharmonic and high frequency conducted disturbances generated by power electronics (e.g. Active Infeed Converters). Compatibility requirements addressing the intentional signals and disturbances are under consideration by EMC technical committees.

4.11 Rapid voltage change

Under normal operating conditions (excluding events), rapid voltage changes should not exceed indicative values.

Rapid voltage change indicative values are in the range of 3~5 % of U_N (U_C) for LV, MV and HV.

These values specifically refer to relative steady-state voltage changes aggregated over very-short time intervals e.g. 150/180-cycle time intervals (all variations during these intervals are to be aggregated in the r.m.s. value of the so-called steady-state voltage). They are based on the usual design criteria for reactive compensation equipment and motor starting, for example.

In some countries/regions, there are no RVC limits specified. Where national/regional circumstances make it appropriate, different limits may be specified than the values given in this Technical Specification.

NOTE In this Technical Specification, no values are given for the maximum voltage change (d_{max}) as defined in IEC 61000-3-3

4.12 Transient overvoltage

4.12.1 Low voltage systems

Transient overvoltages at the supply terminals of LV systems are generally caused by lightning (induced overvoltage) or by switching in the system or in the installation. More information on overvoltage can be found in IEC 61000-2-14.

NOTE 1 The rise time can cover a wide range from milliseconds down to much less than a microsecond. However, for physical reasons, transients of longer durations usually have much lower amplitudes. Therefore, the coincidence of a high amplitude and a long rise time is extremely unlikely.

NOTE 2 The energy content of a transient overvoltage varies considerably according to the origin. An induced overvoltage due to lightning generally has a higher amplitude but lower energy content than an overvoltage caused by switching, because of the generally longer duration of such switching overvoltages.

For withstanding transient overvoltages in the vast majority of cases, where necessary (see IEC 60364-4-44), surge protective devices should be selected according to IEC 60364-5-53, to take account of the actual situations. This is assumed to cover also induced overvoltages due to both lightning and switching.

4.12.2 Medium and High voltage systems

Transient overvoltages in MV or HV supply systems are caused by switching or by lightning, directly or by induction. Switching overvoltages generally are lower in amplitude than lightning overvoltages, but they can have a shorter rise time and/or longer duration. More information on overvoltage can be found in IEC 61000-2-14.

The network users' insulation coordination scheme should be compatible with that adopted by the network operator.

5 Objectives and methods for power quality assessment

5.1 General

Generally, power quality assessments are made for:

- Network operator performance evaluation

This evaluation is often required by either regulatory authorities or by the network users to value power quality against relevant standards, e.g. assessment for survey, complaint, verification of compliance with connection agreement, compliance with quality regulation, or benchmarking.

- Trouble shooting

To diagnose power quality related problems such as system harmonic resonance, custom producing process abnormal interruption, equipment malfunction, etc., typically, raw unaggregated power quality measurement data are most useful for troubleshooting, as they permit any type of post-processing preferred.

In this case, emphasis should be focused on the measurement of current which is invaluable in determining sources/causes of power quality disturbances, since it can help to determine if the cause of the problem is up stream or downstream of the measuring instrument.

- System planning

Power quality is an important aspect for the development of the network for system expansion or connection of new sensitive or disturbing installations (these could be either a load or a generator).

Attention must be paid to the power quality assessment to ensure that the power quality evaluation process addresses the following 5 aspects explicitly:

- recommended or/indicative values;
- system conditions associated with recommended values;
- points on which power quality recommended values are applied;
- methods how power quality parameters are to be measured;
- methods by which assessment results come out from the large number of measurement field data. These methods are provided in Section 5.2 and 5.3 hereafter.

Although some types of disturbances are assumed to be continuous phenomena likely to be present on any supply terminals, it is also important to keep in mind that these types of disturbances vary widely in time. Accordingly, any assessment approach used should carefully consider whether the time varying nature of these disturbances is properly addressed.

5.2 Site power quality assessment

5.2.1 General

For site power quality assessment, two kinds of methods are often used commonly depending on the concerning phenomena.

- Statistical indices like percentile values, maximum or mean values over a certain period of time
- Event counting and tabulating.

5.2.2 For continuous phenomena

For continuous power quality phenomena assessment focusing on site, methods defined in Table 6 are recommended in this Technical Specification.

Table 6 – Site power quality assessment methods

Phenomena	Minimum assessment period ^a	Assessment indices			Indicative value ^d	
		not exceeding corresponding recommended values ^b		not exceeding <i>k</i> times corresponding recommended values ^c		
Supply voltage deviation	1 week	Voltage	Low level	Upper level	—	Max. upper level and low level supply voltage daily 3 s values with corresponding time stamp.
		LV (see 4.3.2)	β % weekly 10 minute r.m.s. values, $\beta = 5$ or $\beta \in \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ [0, 1]	ρ % weekly 10 minute r.m.s. values. $\rho \in [99, 100]$		
			if $\beta > 100 - p$, (100 – p) % weekly 10 minute r.m.s. values.			
		MV (see 4.3.3)	1 % weekly 10 minute r.m.s. values.	99 % weekly 10 minute r.m.s. values		
			0 % weekly 10 minute r.m.s. values.			
		HV (see 4.3.4)	1 % weekly 10 minute r.m.s. values	99 % weekly 10 minute r.m.s. values.		
Voltage unbalance	1 week	95 % weekly 10 minute r.m.s. values (see 4.4)		99 % daily 3 s values	Max. daily 3 s values with corresponding time stamp.	
Flicker	1 week	95 % weekly 2 h P _{lt} values (see 4.5)		99 % daily 10 minute P _{st} values	Max. daily 10 minute P _{st} values with corresponding time stamp.	
Harmonics and inter-harmonics	1 week	95 % weekly 10 minute r.m.s. values (see 4.6)		99 % daily 3 s values	Max. daily 3 s values with corresponding time stamp.	

Phenomena	Minimum assessment period ^a	Assessment indices		Indicative value ^d
		not exceeding corresponding recommended values ^b	not exceeding <i>k</i> times corresponding recommended values ^c	
Mains signalling voltage	1 day	99 % daily 3 s values (see 4.10)	–	Max. daily 3 s values with corresponding time stamp.

^a For long time measurement assessment, an assessed weekly value should be retained on a daily sliding basis; Figure 2 is an example.

^b Assessment for survey, complaint, verification of compliance with connection agreement, compliance with quality regulation, or benchmarking.

^c Assessment for more detailed power quality during assessment period. The coefficient *k* shall be determined by long term measurement campaigns in several sites in order to proper characterize the voltage supplied with the PQ very short-time indices given in Table 6. The detailed value/or range of *k* for each phenomenon is under consideration.

^d Only for trouble shooting purpose. Here corresponding time stamp with appropriate time resolution is needed for further post analysis in case of incident occurring at the same time to find out the relationship between the incident and the power quality parameter indices.

NOTE 1 Flagging concept is used in this Technical Specification according to IEC 61000-4-30. Unless otherwise indicated, e.g. for voltage deviation, the flagged data is excluded in the calculation of percentile values.

NOTE 2 Here in Table 6, 10 minute r.m.s. values are referred to '10 min interval' values defined by IEC 61000-4-30; 3 s values are referred to '150/180-cycle interval' values (150 cycles for 50 Hz nominal or 180 cycles for 60 Hz nominal) defined by IEC 61000-4-30.

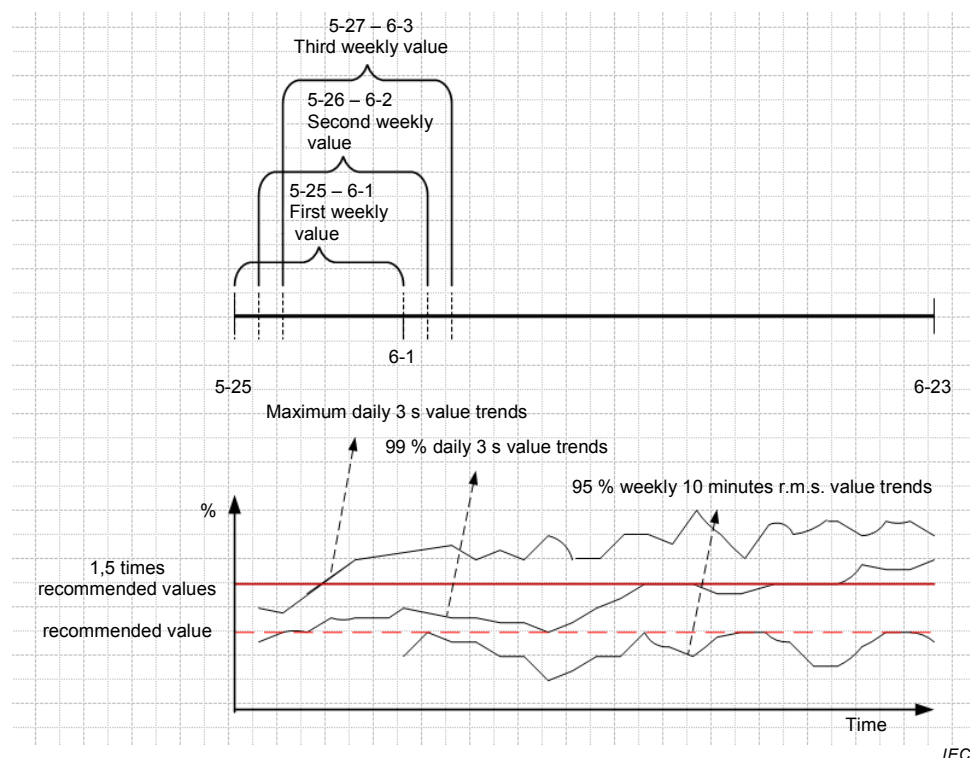


Figure 2 – An example for illustrating voltage THD assessment result trends

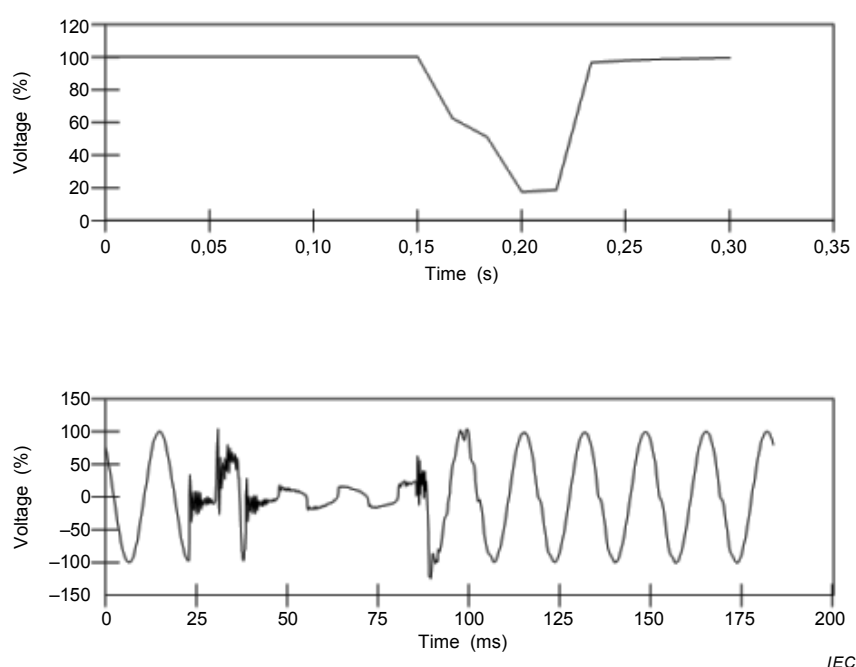
5.2.3 For discontinuous phenomena (single event)

For individual event evaluation, residual voltage and duration combined with the RMS voltage variation shape during each event can be tabulated like in the example for a single event assessment presented in Table 7 and Figure 3.

NOTE The measurement and detection methods for voltage swells, dips and short time interruptions should be in accordance with IEC 61000-4-30.

Table 7 – Example of single event assessment

Event Attribution	Detailed Characterization
Location	East station 10 kV busbar
Time Stamp	2011-06-30 12 h:36 m:12.2150 s
Capturing Threshold	80 %
Residual Voltage	21 %
Time Duration	81,9 ms
RMS Variation Shape	Upper part in Fig 3
Point on Wave	Lower part in Fig 3

**Figure 3 – An example showing information of single event assessment**

5.3 System aspect power quality assessment

5.3.1 General

For system (sub-system/area) power quality assessment, weighting rules may be used applying both to statistical indices and events in order to get global results depending on the details of collection of single points.

5.3.2 For continuous phenomena

For system power quality indices, there are no corresponding recommended values, but the assessment based on site evaluation results can give very useful information for system power quality management.

Annex B gives an example of system aspect continuous disturbance evaluation.

5.3.3 For discontinuous phenomena (events)

5.3.3.1 General

For system aspect assessment of events, the following SARFI methods combining with the magnitude-duration table (TABLE 10) can be used.

NOTE 1 Detailed information about SARFI methods is presented in IEEE 1564.

NOTE 2 For system aspect assessment of events, several methods are also presented in CENELEC TR 50555, *Interruption indices*.

Time aggregation method, in the case of multiple successive events, should be used prior to the assessment. The time aggregation duration is defined as 1 minute in this Technical Specification, within which all events can be counted as one event whose magnitude and duration are those of the most severe observed during this interval.

Different aggregation methods may be selected for individual purposes; some reference rules are given in IEC TR 61000-2-8.

5.3.3.2 SARFI method

SARFI is an acronym for System Average RMS variation Frequency Index (SARFI). It is a power quality index that provides a count or rate of voltage dips, swells, and/or interruptions for a system. The size of the system is scalable: It can be defined as a single monitoring location, a single network user service, a feeder, a substation, groups of substations, or for an entire power delivery system.

SARFI-X corresponds to a count or rate of voltage dips, swells and/or interruptions below or above a voltage threshold. For example, SARFI-70 considers voltage dips and interruptions that are below 0,70 per unit, or 70 % of the reference voltage. SARFI-110 considers voltage swells that are above 1,1 per unit, or 110 % of the reference voltage.

In this Technical Specification, a rate of 30 days (number of events/30 days) is recommended for assessment of voltage dips, swells and/or interruptions below or above a voltage threshold.

The following is an example using the SARFI-X method. Table 8 presents a list of residual voltages with corresponding events duration at a single monitoring site and Table 9 gives SARFI-X indices coming out of Table 8. The observation period is from Jul-01-2000 to Oct-1-2000, total 92 days.

Table 8 – List of individual events measured at a single monitoring site

Time Stamp	Residual Voltage (%)	Event Duration (ms)
Jul-01-2000 09:48:52	73	180
Jul-01-2000 09:50:16	73	180
Jul-07-2000 14:20:12	0	1640
Jul-10-2000 15:55:23	13	2000
Jul-21-2000 09:48:52	0	2600
Aug-08-2000 07:35:02	49	680
Sep-02-2000 08:30:28	0	41000
Sep-08-2000 10:30:40	59	800

Table 9 – SARFI-X indices coming out of Table 8

Index	Count	Events per 30 Days
SARFI-90	8	2,61
SARFI-70	6	1,96
SARFI-50	5	1,63
SARFI-10	3	0,98

5.3.3.3 Magnitude-duration table

It is clear that when using the SARFI method, the event time duration information is missing. This will be remedied by using the magnitude-duration table described hereafter.

Magnitude – duration table format is shown as Table 10. Each cell populates the total count summing each event matching the corresponding residual voltage and duration restriction. The values in Table 10 are out of Table 8.

Table 10 – Magnitude – duration table format

Residual voltage $U\%$	Duration t (ms)				
	$10 \leq t \leq 200$	$200 < t \leq 500$	$500 < t \leq 1\,000$	$1\,000 < t \leq 5\,000$	$5\,000 < t \leq 60\,000$
$U \geq 120$	0	0	0	0	0
$120 > U > 110$	0	0	0	0	0
$90 > U \geq 80$	0	0	0	0	0
$80 > U \geq 70$	2	0	0	0	0
$70 > U \geq 40$	0	0	2	0	0
$40 > U \geq U_{\text{ith}}$	0	0	0	1	0
$U_{\text{ith}} > U$					
Voltage interruption	0	0	0	2	1

NOTE On a three phase system, voltage interruption begins when the U_{rms} voltages of all three phases fall below the interruption threshold (U_{ith}). Refer to IEC 61000-4-30 for more information.

Annex A (informative)

Examples of PROFILES for Power Quality Specification

The following information is provided by experts from different countries/regions as examples.

NOTE This annex is intended to address the flexibilities offered in this TS, not to identify all the differences with applicable national/regional standards.

A.1 LV public distribution in European countries

LV public distribution in European countries (applicable standard: EN 50160)	
4.2	For systems with synchronous connection to an interconnected system: – 50 Hz $\pm$ 1 % during 99,5 % of a year; – 50 Hz + 4 % / – 6 % during 100 % of the time. For systems with no synchronous connection to an interconnected system (e.g. supply systems on certain islands): – 50 Hz $\pm$ 2 % during 95 % of a week; – 50 Hz $\pm$ 15 % during 100 % of the time. NOTE The frequency assessment is based on 10s values according to IEC 61000-4-30.
4.3.2	$\rho = 100$ $\beta = 5$
4.4	–
4.5	–
4.6.2.1	THD is calculated with harmonic orders up to 40th only NOTE U_{19}, U_4 and $U_{6...24}$ in Table 3: see EN 50160 (Table 1)
4.6.2.2	–
4.7	–
4.8	–
4.9	Interruptions threshold = 5 % of reference voltage
4.10	–
4.11	No rapid voltage change limits are specified
4.12.1	–

A.2 LV, MV and HV power supply system in China

LV,MV and HV power supply system in China																												
4.2	For LV, MV, and HV power supply systems with synchronous connection to an interconnected system: – 50 Hz ± 0,2 Hz For systems with no synchronous connection to an interconnected system or for weaker systems: – 50 Hz ± 0,5 Hz																											
4.3	ρ and β: not defined For LV power supply systems: – $U_N \pm 7\%$ For MV and HV power supply systems: – abs (plus deviation) + abs (minus deviation): 10 % (abs: Absolute Value Sign)																											
4.4	–																											
4.5	–																											
4.6.2.1, 4.6.3.1 and 4.6.4.1	– THD is calculated with harmonic orders up to 25th only – Harmonics for LV,MV, and HV power supply systems: <table><tr><th rowspan="2">Voltage (kV)</th><th rowspan="2">THD (%)</th><th colspan="2">HR (%)</th></tr><tr><th>Odd harmonics</th><th>Even harmonics</th></tr><tr><td>0,38</td><td>5,0</td><td>4,0</td><td>2,0</td></tr><tr><td>6</td><td rowspan="2">4,0</td><td rowspan="2">3,2</td><td rowspan="2">1,6</td></tr><tr><td>10</td></tr><tr><td>35</td><td rowspan="2">3,0</td><td rowspan="2">2,4</td><td rowspan="2">1,2</td></tr><tr><td>66</td></tr><tr><td>110</td><td>2,0</td><td>1,6</td><td>0,8</td></tr></table>				Voltage (kV)	THD (%)	HR (%)		Odd harmonics	Even harmonics	0,38	5,0	4,0	2,0	6	4,0	3,2	1,6	10	35	3,0	2,4	1,2	66	110	2,0	1,6	0,8
Voltage (kV)	THD (%)	HR (%)																										
		Odd harmonics	Even harmonics																									
0,38	5,0	4,0	2,0																									
6	4,0	3,2	1,6																									
10																												
35	3,0	2,4	1,2																									
66																												
110	2,0	1,6	0,8																									
4.6.2.2, 4.6.3.2 and 4.6.4.2	interharmonics for LV,MV, and HV power supply systems: <table><tr><th>Voltage</th><th><100 Hz</th><th>100 ~ 800 Hz</th></tr><tr><td>$U_N \leq 1000V$</td><td>0,2</td><td>0,5</td></tr><tr><td>$U_N > 1000V$</td><td>0,16</td><td>0,4</td></tr></table> Note Value here is for interharmonic ratio.				Voltage	<100 Hz	100 ~ 800 Hz	$U_N \leq 1000V$	0,2	0,5	$U_N > 1000V$	0,16	0,4															
Voltage	<100 Hz	100 ~ 800 Hz																										
$U_N \leq 1000V$	0,2	0,5																										
$U_N > 1000V$	0,16	0,4																										
4.7	–																											
4.8	–																											
4.9	–																											
4.10	No Mains signalling voltage limits are specified																											
4.11	No rapid voltage change limits are specified																											
4.12	See GB/T 18481																											

A.3 Example of a transmission system in Canada

Applies to high voltage 60 Hz networks from 44 kV to 230 kV				
4.2	60 Hz ± 1% (i.e. 59,4 to 60,6 Hz), during 99,9% of a year; NOTE Does not apply to islanded systems.			
4.3.4	–			
4.4	Negative voltage unbalance factor is: – 1,5 % for nominal voltage 230 kV; – 2 % for nominal voltage from 44 kV to 161 kV.			
4.5	P _{lt} = 0,8			
4.6.4.1	Table 5 – Indicatives values of individual harmonic voltages at the high voltage supply terminals given in percent of the fundamental voltage U_1			
	Odd harmonics		Even harmonics	
	Order	$U_{h95\%}$	Order	$U_{h95\%}$
	h	(%)	h	(%)
	3	2	2	1,5
	5	2	4	1
	7	2	6 ≤ h ≤ 50	0,5
	9	1,5		
	11	1,5		
	13	1,5		
	15	1		
	17 ≤ h ≤ 49	1,2 × 17/h		
Voltage THD = 3 %				
4.6.4.2	Voltage THDG = 3 %			
4.7	–			
4.8	–			
4.9	Interruption threshold is 10 % of reference voltage.			
4.10	n/a			
4.11	RVC limit = 3 %. In some circumstances*, it can reach 6 % of nominal voltage. * These are viewed as degraded operating conditions when equipment switching must be carried out to meet supply system or load requirements.			
4.12.2	–			

A.4 Examples of Profiles in Australia

Examples of Profiles in Australia																										
4.2	For the Australian mainland under normal conditions with no contingency or load event, the required frequency is: – 49,75 to 50,25 Hz, while maintaining 49,85 to 50,15 Hz 99 % of the time over a 30 day period. – For islanded systems, under normal conditions with no contingency or load event, the required frequency is: 49,5 to 50,5 Hz – Compliance is based on 4 second measurements																									
4.3.2	Based on what is published in AS 61000.3.100, it specifies V99% and V1%. However, this standard has not yet had widespread adoption by the various state regulators. Accordingly, a wide range of practices currently still be adopted. In any case according to AS61000.3.100 the parameters are: – $\rho = 99$ – $\beta = 1$																									
4.4	Voltage unbalance is 2 % 30 minute values for 100 % of the time																									
4.5	– P_{lt} for LV/MV is 1,0 for 95 % of the time – and P_{lt} for HV is 0,8 for 95 % of the time																									
4.6.2.1	Under consideration – however Victorian Distribution code applies IEEE Standard 519-1992 , and total harmonic distortion limit of 5 %																									
4.6.2.2	As per IEC 61000-2-2																									
4.7	Preferred 230V voltage dip and swell measurement thresholds <table><tr><th rowspan="2">Voltage threshold</th><th colspan="2">Phase-to-neutral voltage</th><th colspan="2">Phase-to-phase voltage</th><th colspan="2">1 phase 3 wire centre Neutral phase-to-phase voltage</th></tr><tr><th>dip</th><th>swell</th><th>dip</th><th>swell</th><th>dip</th><th>swell</th></tr><tr><td>1/2 cycle r.m.s.</td><td>207 V</td><td>262 V</td><td>360 V</td><td>456 V</td><td>414 V</td><td>524 V</td></tr></table>						Voltage threshold	Phase-to-neutral voltage		Phase-to-phase voltage		1 phase 3 wire centre Neutral phase-to-phase voltage		dip	swell	dip	swell	dip	swell	1/2 cycle r.m.s.	207 V	262 V	360 V	456 V	414 V	524 V
Voltage threshold	Phase-to-neutral voltage		Phase-to-phase voltage		1 phase 3 wire centre Neutral phase-to-phase voltage																					
	dip	swell	dip	swell	dip	swell																				
1/2 cycle r.m.s.	207 V	262 V	360 V	456 V	414 V	524 V																				
4.8	See 4.7 above																									
4.9	(a) for single phase systems, falls below the interruption threshold of 10 % of the nominal voltage, U_n for at least a ½ cycle (b) for poly-phase systems, falls below the interruption threshold of 10 % of the nominal voltage, U_n for at least a ½ cycle on all channels																									
4.10	Presently as per IEC																									
4.11	No limits specified other than via flicker requirements																									
4.12	No limits specified																									

Annex B (informative)

Example on System Aspect Continuous Disturbance Evaluation

Firstly, two variables named Average Index $\bar{x}$ and Sample Standard Deviation s are defined by equations at B1 and B2.

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} \quad (B1)$$

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}} \quad (B2)$$

$\bar{x}$ Site average value of one kind of disturbances, e.g., unbalance, flicker etc.

x_i The assessment result (percentile values) of individual site belonging to this system (area).

n Total number of monitoring sites covered in this system (area).

s Sample standard deviation.

Thus, Average Index $\bar{x}$ and Sample Standard Deviation s are used for system (sub-system, or area) aspect power quality evaluation. Table B.1 is an example of the results using this method.

Table B.1 – Listing of System Power Quality Evaluation

Disturbance		Average value	Sample standard deviation	Maximum of site value	Minimum of site value
Voltage THD (%)					
Unbalance (%)					
Flicker	P_{It}				
NOTE The total number of sites should be list here.					

NOTE Max. and Min. site values are based on site percentile values.

Annex C

(informative)

Main Impact of Poor Power Quality

When supplied by AC electricity, voltage magnitude and frequency are always the key factors of concern for power supplies and end users as these factors are indeed fundamental to the supply of quality power. If the supplied voltage or frequency is not within the reasonable range, the performance of network users' equipments will be impacted, and so will be the power system itself.

Power quality indices are described the characteristics of supplied power in order to clarify the responsibility between the system operators and the end users. Power utilities should shoulder responsibility for maintaining the reasonable range of power quality levels. On the other hand, power quality also depends on the way in which it is being used at any instant by the equipment of multiple users. Thus maintaining the preferred power quality condition is the responsibility of both the system operators and end users, namely through the enforcement of standards for controlling emission limits and ensuring a minimum immunity of end-use equipment.

C.1 Harmonic distortion

The long time exposure to relatively high harmonic distortion conditions may cause some serious effects on the various pieces of equipments. Even very high short-term harmonics distortion, e.g. resonance condition, may cause dielectric breakdown due to over voltages.

Harmonics can lead to overloading. Hence overheating increases dielectric stress and the power loss.

- Capacitors for power factor correction often act as sinks for a particular order of harmonic currents. In this case, it can lead to capacitor over current if no forethought is given at the designing stage.
- Non-sinusoidal power supplies result in reduction of torque of induction motors.
- Harmonics will increase interference with telephone, communication and analogue circuits.
- Excessive levels of harmonics can cause errors in reading of induction type energy meters which are calibrated for pure sinusoidal AC power.
- High order harmonics cause voltage stresses.
- Harmonic currents flowing through power system networks can cause additional losses.

It is reported that the level of interharmonics in power supply systems is increasing due to the development of frequency converters and similar electronically controlled equipment. HARMONIC VOLTAGES and INTERHARMONIC VOLTAGES, if not controlled, might lead (among other effects) to overloading or disturbance of equipment on the supply networks and in electricity users' installations.

In some cases, INTERHARMONIC VOLTAGES, even at low levels, can give rise to flicker, or cause interference in ripple control systems.

C.2 Voltage unbalance

Voltage unbalance is always a concern as it affects the transformers, electrical motors, and electrical generators.

- Voltage unbalance degrades the performance and shortens the life of a three-phase motor.
- Current unbalance caused by voltage unbalance essentially creates counter-torque (resistive torque). That is, it tries to make the motor turn in the opposition direction. This will create heating.
- Voltage unbalance may also reduce the capacity of equipment such as motors or generators if not properly taken into consideration at the design stage (equipment is

normally designed and rated to account for some degree of voltage unbalance normally present in any power system).

C.3 Voltage deviation

Large voltage deviations from the nominal values will shorten the life of electrical equipment, lower the stable limit of power system and increase the cost of network operation. Equipment operating under this condition will be malfunction, breakdown or damaged.

C.4 Frequency deviation

If frequency deviation exceeds the limit, the motor shall be protected by means stopping the operation.

C.5 Voltage fluctuation

Voltage fluctuations can cause a number of harmful technical effects such as data errors, memory loss, equipment shutdown, flicker, motors stalling and reduced motor life, resulting in disruption to production processes and substantial costs. However considering the fact that voltage fluctuations are normally within 10 % magnitude, most of these above mentioned effects are more typical of voltage dips or swells.

C.6 Flicker

Flicker is considered to be an annoying problem for the NETWORK USERS. Most of the time, it does not have a high financial impact. However, at high levels it can cause inconvenience to people when frequent flickering of lights (Different technology of lamps may have different sensitivity to voltage fluctuation.) and computer screens occurs at their work-places or homes.

C.7 Voltage dip (or Voltage sag)

Motor drives, including variable speed drives, are particularly susceptible because the load still requires energy that is no longer available except from the inertia of the drive. In processes where several drives are involved, individual motor control units may sense the loss of voltage and shut down the drive at a different voltage level from its peers and at a different rate of deceleration, resulting in complete loss of process control. Data processing and control equipment is also very sensitive to voltage dips and can suffer from data loss and extended downtime.

C.8 Transient overvoltages

Transient overvoltages can cause large dV/dt values that can damage or reduce the lifetime of variable speed drives.

Annex D

(informative)

Power Quality Issues Related to Distributed Generation and Micro-grids

Because of grid connected Distributed Generation (DG), the distribution network will eventually change from transmitting and allocating electrical energy to a new power exchange system, which includes power collection, transmission and storage. Consequently it brings a series of problems including power quality issues.

DG is also the main power element making up micro-grids. The following PQ characterizations are always present in micro-grids.

D.1 Voltage deviation

Grid connected DG changes the distribution network power flow and can cause reverse power flow. This affects distribution network voltage and may cause large voltage deviations. The deviation mainly depends on the network capability and location of grid connected DG. The closer the DG's location to the supply terminal node, the bigger the voltage change would be at the supply terminals, and vice versa. Thus some node voltages will be higher than the rated voltage or the bus line voltage because of improper application of the DG.

D.2 Harmonics

DGs such as photovoltaic cells (PV), wind power, fuel cells, power storage systems and gas turbines can be connected to power system by power electronics devices.

- PV, fuel cells and storage systems generate direct current which needs an inverter to connect to the power system.
- Gas turbine generates high frequency alternating current, which needs AC/DC/AC or AC/AC frequency converters to connect to the power system.
- Advanced wind power connects to the power system by an AC/DC/AC converter, transforms AC voltage into DC voltage and then transforms the DC voltage into rated frequency AC voltage.

These devices convert electric power, control load and may cause grid voltage and current wave distortion leading to similar effects to that of adding non-linear load to the power system.

D.3 DG magnetic bias (DC current injection)

In a DG system which connects to the grid through inverter, if the parameters are unbalanced or trigger impulses that are asymmetrical, there may appear to be DC current in the inverters. Inflow of the DC current in the distribution transformers may cause the transformer's DC magnetic bias and consequently cause wave distortion and abnormal heating as a result. In order to attenuate this phenomenon, isolation transformers can be used, but the root problem of filtering and avoiding unequal firing should be corrected in the first place.

D.4 Voltage fluctuation and flicker

For renewable power generation systems like wind and photovoltaic, the unpredictable fluctuating source of electricity is the main reason leading to output electric power fluctuation. It can be increased by the use of maximum power point tracking (MPPT) control technology for improvement of efficiency of a DG system.

On the other hand, DG system operation is usually controlled by its property owner. This may cause DG system to randomly start-up and stop operation. When some large capacity DG systems start or quit, the power output changes abruptly causing voltage fluctuations and flicker. This is more serious when the short-circuit capacity of the DG connection point is low.

D.5 High frequency conducted disturbances

Moreover, grid connected active infeed converters may be a source of high frequency conducted disturbances (e.g. in the range 2 kHz – 150 kHz) and interference cases with electronic devices, electricity meters and PLC systems for smart meter/grid have been reported.

Annex E (informative)

Methods to Maintain and Improve Power Quality

The quality indices and the recommended values cover appropriately the vast majority of locations under acceptable economic conditions, despite the differences in situations, provided that:

- For mass-market products, emission requirements in standards such as IEC 61000-3-2, 3-3, 3-11 and/or 3-12, are regularly and appropriately updated to take into account the development of markets and changes in technologies;
- For large installations, emission levels are effectively controlled, e.g. through connection agreements;
- Network operators make use of appropriate methodologies and engineering practices, e.g. based on PLANNING LEVELS and IEC TR 61000-3-6, 3-7, 3-13 and/or 3-14.

E.1 Voltage deviation

The deviation of supply voltage is mainly caused by voltage drops as a result of load current flowing through system internal impedance.

Generally, voltage deviation requirements controlling operations are made by system operators, including:

- Voltage regulation, including system aspects of active power flow control and reactive power flow optimizing, transformer taps adjusting, etc.
- Reactive power compensation at distribution stations, including capacitor banks and reactors equipments, which are switched automatically according to the connected voltage.

On the other hand, the end users are required to comply with the power factor demand regulations made by (agreed with) system operators. Thus reducing the reactive current draw from the supply system with reactive power compensation equipment is the main method for end users to meet voltage deviation requirements.

For large fluctuating end users, the following advanced equipment may be best choice technically and economically.

- Static VAR Compensator (SVC);
- Magnetic Controlling Reactors (MCR);
- Static Synchronous Compensators (STATCOM) or Static VAR Generation (SVG).

E.2 Harmonics

Mitigating harmonics for the NETWORK USER begins at the disturbance source. The following process may be the choice according to the particular circumstances.

- Passive Filters (LC filtering circuits)
- This is the preferred industry choice as the equipment acts as the sink for certain harmonic current orders.
- Active Power Filters (APF)
- Mostly used for harmonic current fluctuating situations, thus the response time is the key factor for characterizing its performance.
- Embedded solution.
- Eg., PWM (Pulse Width Modulation) technology used in modern power electronic communities.

E.3 Flicker

Similar with harmonics, mitigating flicker for the NETWORK USER begins at the disturbance source. It is always accomplished by controlling fluctuating power drawn by the disturbance load, e.g., electric arc furnace and elevator.

- Use of higher voltage level supply as agreed between system operators and end users
- Static Var Compensators (SVC)
- Static Synchronous Compensators (STATCOM) or Static Var Generation (SVG)

In cases where SVCs, STATCOMs, or SVGs are used, response time is the key factor for characterizing its performance

E.4 Unbalance

To decrease the degree of unbalance, several actions can be taken:

- The first and most basic solution is for the system operators to deploy or distribute the loads in such a way that the three phase loads become more balanced;
- For large unbalanced loads, the SVC technology has the result according to C.P.Steinmetz theory;
- For some of the railway applications powered by single phase supplies, special transformers, such as Scott and Steinmetz transformers, are to be used;
- In a three phase system, transposition of transmission lines may also be a solution to attenuate the negative sequence unbalance voltage.

E.5 Voltage dip/swell/short time interruption

Voltage dips, swells and short time interruptions are important PQ problems affecting industrial and large commercial NETWORK USERS. As most of these events are caused by circuit faults, improving system operation management skills and constructing robust supply systems are always the fundamental procedure to decrease the effects of these unpredictable events.

Another efficient method is an embedded solution to improve the sensitive load immunity for riding through these events. For example, by the application of devices such as:

- Uninterruptible power supplies (UPS's);
- Energy storage devices (Super Capacitors, SMES, etc.);
- And dynamic Voltage Restorers(DVR).

Annex F (informative)

Relation between Power quality and EMC

Power quality is related to EMC in a number of ways – especially because compliance with power quality requirements depends on the control of cumulative effect of electromagnetic emissions from all/multiple equipment and/or installations.

The values recommended in this Technical Specification for power quality purposes are equal or very close to the compatibility levels in EMC standards (e.g. IEC 61000 series) but, with a somewhat different meaning. In particular, they differ with respect to their point of application and probability of being exceeded.

According to IEC 61000-2-2 and IEC 61000-2-12, a compatibility level is a specified electromagnetic disturbance level used as a reference level in a specified environment for coordination in the setting of emission and immunity limits.

By convention, the compatibility level is chosen so that there is only a small probability that it will be exceeded by the actual disturbance level. According to IEC TR 61000-3-6, compatibility levels are generally based on the 95 % probability levels of entire systems, using statistical distributions which represent both time and space variations of disturbances. There is allowance for the fact that the network operator cannot control all points of a system at all times. Therefore, evaluation with respect to compatibility levels is made on a system-wide basis and not at specific locations.

This TS defines the characteristics of electricity at the supply terminals of public networks. So the recommended values for power quality indices apply at any point of public networks. Moreover, all weekly 95% values should meet the recommended values. So, even if the power quality recommended values in this TS are equal or very close to the compatibility levels defined in IEC 61000-2-2 and IEC 61000-2-12 for unbalance, harmonics and interharmonics, the criteria are far more severe.

NOTE These differences can make power quality recommended values difficult to meet, in some situations

While the power quality requirements in this Technical Specification apply only to the SUPPLY TERMINALS, these requirements should be coordinated with those concerning the electromagnetic environment of electrical equipment, the utilization point of equipment and the point of common coupling.

The change in voltage characteristics between the network user's supply terminals and the equipment terminals/utilization point depends on installation rules and is impacted differently by particular phenomena/ electromagnetic disturbances.

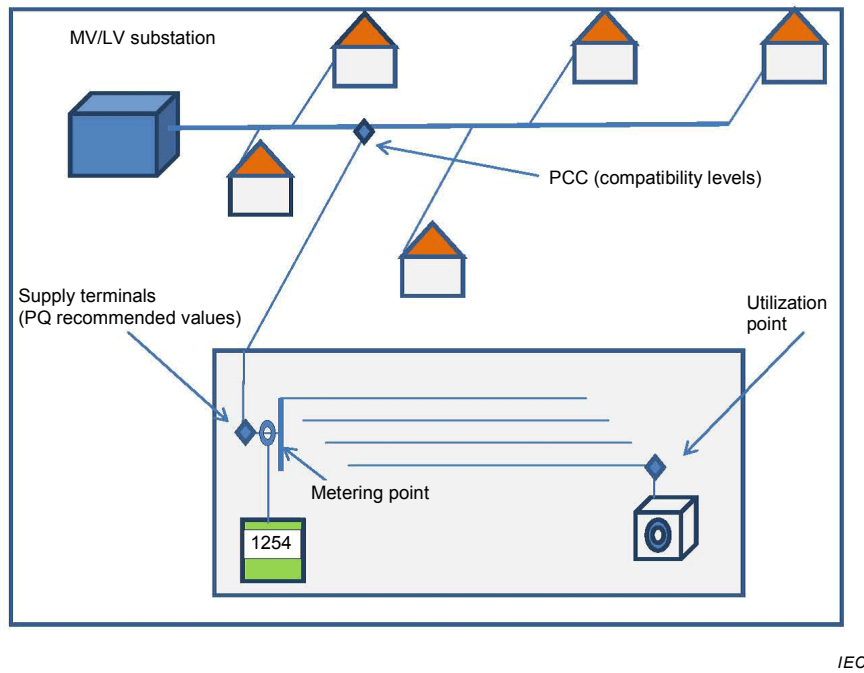


Figure F.1 – Application points in a LV system (example)

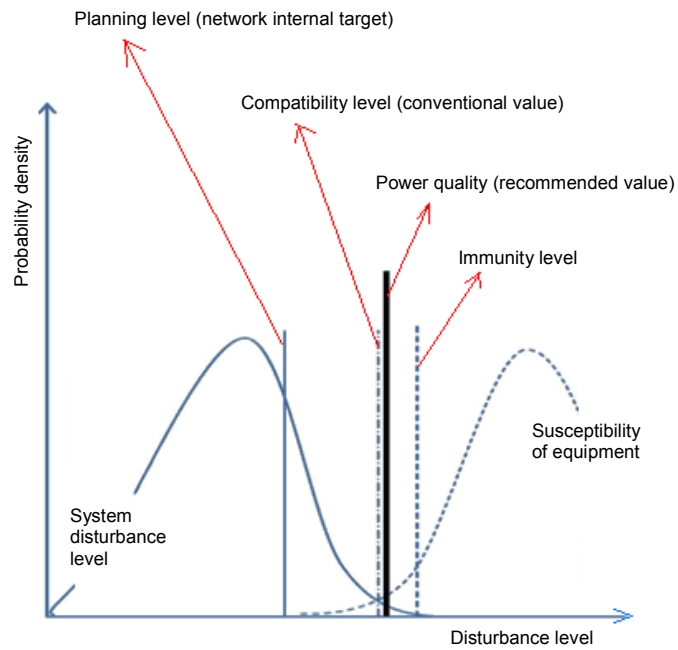


Figure F.2 – Relation between disturbance levels (Schematic significance only)

Bibliography

- [1] IEC TR 62510, Standardizing the characteristics of electricity
 - [2] IEC TR 61000-2-1, Electromagnetic compatibility (EMC) – Environment – Description of the environment – Electromagnetic environment for low frequency conducted disturbances and signalling in public power supply systems
 - [3] IEC 61000-2-4, Electromagnetic compatibility (EMC) – Environment – Compatibility levels in industrial plants for low-frequency conducted disturbances
 - [4] IEC TR 61000-2-5:2011, Electromagnetic Compatibility (EMC) Environment. Description and classification of electromagnetic environments
 - [5] IEC TR 60725, Consideration of reference impedances and public supply network impedances for use in determining disturbance characteristics of electrical equipment having a rated current = < 75 A per phase
 - [6] EN 50160, Voltage characteristics of electricity supplied by public distribution networks
 - [7] CENELEC TR 50422, Guide for the application of the European Standard EN 50160
 - [8] CENELEC TR 50555, Interruption indices
 - [9] AS 61000.3.100 Australian Standard – Electromagnetic compatibility (EMC) Part 3.100: Limits—Steady state voltage limits in public electricity systems
 - [10] IEEE 519, Recommended practices and requirements for harmonic control in electric power systems
 - [11] IEEE 1564-2014, IEEE Guide for Voltage Sag Indices
 - [12] CEER Benchmarking Reports on the Quality of Electricity Supply
 - [13] NRS 048-2:2003, ELECTRICITY SUPPLY — QUALITY OF SUPPLY Part 2: Voltage characteristics, compatibility levels, limits and assessment methods.
 - [14] CIGRE Technical Brochure 261, Power Quality Indices and Objectives, Working Group C4.07, October 2004
 - [15] GB/T 18481, Power quality – Temporary and transient overvoltages
-

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

3, rue de Varembé
PO Box 131
CH-1211 Geneva 20
Switzerland

Tel: + 41 22 919 02 11
Fax: + 41 22 919 03 00
info@iec.ch
www.iec.ch